Single-Layer Perceptron – Binary Classification

Project Overview

This project implements a **Single-Layer Perceptron using TensorFlow and Keras** to perform binary classification on the Breast Cancer Wisconsin dataset. The model is trained for 25 epochs, evaluated using classification metrics, and presented through an interactive website dashboard built with Gradio.

Objectives

- Prepare a dataset for binary classification.
- Design and train a single-layer perceptron.
- Understand activation functions, loss functions, optimizers, and epochs.
- Evaluate model performance using accuracy and classification metrics.
- Visualize training and validation performance.
- Display the results through a web-based dashboard.

Technologies Used

- Python
- Google Colab
- TensorFlow
- Keras
- NumPy
- Matplotlib
- Scikit-learn
- Gradio
- Pandas

Dataset

Dataset: Breast Cancer Wisconsin Dataset

The dataset contains 569 samples and 30 numerical features.

The target classes are:

- **Malignant (0):** Cancerous tumour.
- **Benign (1):** Non-cancerous tumour.

The dataset is split into training and testing sets using an 80:20 ratio. StandardScaler is used to standardize the input features.

Model Architecture

The model contains a single Dense output layer with one neuron and a sigmoid activation function.

- **Input:** 30 features
- **Output:** One neuron
- **Activation function:** Sigmoid
- **Loss function:** Binary cross-entropy
- **Optimizer:** Adam
- **Epochs:** 25
- **Batch size:** 32

Key Concepts

Activation Function

The sigmoid function converts the model output into a value between 0 and 1, which can be interpreted as a probability for binary classification.

Loss Function

Binary cross-entropy measures the difference between predicted probabilities and actual labels.

Optimizer

Adam updates the model's weights and bias to reduce the loss during training.

Epochs

One epoch represents one complete pass through the training dataset. This model is trained for 25 epochs.

Model Evaluation

The model is evaluated on the held-out test dataset using:

- Test accuracy
- Precision
- Recall

- F1-score
- Confusion matrix
- Training and validation accuracy
- Training and validation loss

Observed Results

Based on the model's evaluation output:

- **Test accuracy:** Approximately 86.84%
- **Correct predictions:** 99 out of 114
- **Incorrect predictions:** 15 out of 114

Confusion Matrix

Actual Class	Predicted Malignant	Predicted Benign
Malignant	39	3
Benign	12	60

These results describe this particular training run. Results may vary if the dataset split, preprocessing, or model settings change.

Website Dashboard

The project includes a Gradio website dashboard that displays:

- Test accuracy
- Correct and incorrect predictions
- Classification report
- Confusion matrix visualization
- Training and validation accuracy graphs
- Training and validation loss graphs

The dashboard can be launched from Google Colab using Gradio's temporary public sharing URL.

How to Run the Project

1. Open Google Colab.
2. Create a new notebook.

3. Import the required libraries.
4. Load and preprocess the Breast Cancer Wisconsin dataset.
5. Create the single-layer perceptron model.
6. Compile the model using Adam and binary cross-entropy.
7. Train the model for 25 epochs.
8. Evaluate the model and generate the performance plots.
9. Install Gradio and run the dashboard code.
10. Open the generated website URL to view the results.

Expected Outcome

The project demonstrates how a basic neural network can perform binary classification, how to measure its performance, and how to present machine learning results through an interactive web interface.

Conclusion

This assignment provides practical experience with neural network fundamentals, binary classification, model evaluation, and web-based visualization. It connects the machine learning workflow with an interactive dashboard that makes results easier to understand.

Note: This project is intended for educational purposes only. It is not a clinically validated medical diagnostic system.